

Hertzbleed: Turning Power Side-Channel Attacks Into Remote Timing Attacks on x86

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Hertzbleed: Turning Power Side-Channel Attacks Into Remote Timing Attacks on x86

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Power side-channel attacks exploit data-dependent variations in a CPU's power consumption to leak secrets. In this paper, we show that on modern Intel (and AMD) x86 CPUs, power side-channel attacks can be turned into timing attacks that can be mounted without access to any power measurement interface. Our discovery is enabled by dynamic voltage and frequency scaling (DVFS). We find that, under certain circumstances, DVFS-induced variations in CPU frequency depend on the current power consumption (and hence, data) at the granularity of milliseconds. Making matters worse, these variations can be observed by a remote attacker, since frequency differences translate to wall time differences!

The frequency side channel is theoretically more powerful than the software side channels considered in cryptographic engineering practice today, but it is difficult to exploit because it has a coarse granularity. Yet, we show that this new channel is a real threat to the security of cryptographic software. First, we reverse engineer the dependency between data, power, and frequency on a modern x86 CPU—finding, among other things, that differences as seemingly minute as a set bit's position in a word can be distinguished through frequency changes. Second, we describe a novel chosen-ciphertext attack against (constant-time implementations of) SIKE, a

post-quantum key encapsulation mechanism, that amplifies a single key-bit guess into many thousands of high- or low-power operations, allowing full key extraction via remote timing.

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